



Artificial intelligence and robots like us (surgeons) for people like you (patients): toward a new human–robot-surgery shared experience. What is the moral and legal status of robots and surgeons in the operating room?

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Introduction

Artificial intelligence (AI) is an interdisciplinary mathematical topic with algorithms allowing machines to execute pseudo-cognitive activities. AI has a huge impact on medical production. AI can classify pathologies and can predict treatment recommendations. It can be a diagnostic and decision-making tool. Like humans, artificial intelligence builds a shared history through the repetition of interactions. Errors, mismatches, and relational reparations similarly mark these interactions. As a result, a relational memory is formed, and actions are produced in response to the relational experiences. The article of Bouthors et al. [1] describes a step-by-step simulation teaching to improve the student's memory as in artificial intelligence.

In contrast to AI, robots today are simple executors of our commands. The robots' function is related to their sensor and developmental level. The sensor and developmental stage of the robots affect how they work. Their acceptance in various contexts is related to the “like-me” aspect of robots; for example, the guide tour robot in museums, the robot assisting sales, or the robot receptionist, and now the Robot “like-me as a surgeon.” The interaction in orthopaedic surgery is by definition, cursory and bound to the specific domain of bone cutting [2] or sculpting [3] for the knee, reaming for the hip [4], screwing [5], or simply positioning [6].

However, AI and robotics are not for the surgeon's benefit but for the patient's benefit. Lonner et al. [7] propose an “intelligence application” that integrates through a management platform a dialog between care and clinical data obtained from the smartphone of the patient and the data of a commercial robotic knee system. In envisaging the future of human–robot relationship, this article establishes the first form of intersubjectivity to share experiences between humans and machines. Future robots should gradually develop into autonomous collaborating agents rather than mere executors of our explicit commands [8].

The current special issue provides a broad introduction with a condensed overview of artificial intelligence (AI), and robotics, as well as additional comments on future possibilities in orthopaedic surgery.

Increased interest over time in robots

Unicompartmental arthroplasty

The early robots were used for unicompartmental arthroplasty (UKA). Early research on unicompartmental arthroplasty revealed that 100% of the robotically performed group had improved tibiofemoral alignment in the coronal plane within 2° of the desired position compared to only 40% of the manual group [9]. During UKA, robots' technology also improves the position [10] of implants and the possibility of reconstruction of the anterior cruciate ligament [11] when it is absent. Much of the published robotic literature is on partial knee arthroplasty [9–11].

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Total knee arthroplasty

Robotic systems have now been introduced to total knee to improve alignment and functional outcomes in knee replacement surgery. Whether the robotic system is image-based (CT, MRI, or X-ray), or entirely image-free [12], the systems rely completely on the precursor technology of computer-assisted navigation. These images, along with the on-table registration of bone landmarks, enable the construction of an accurate, three-dimensional (3D) virtual model of the knee during surgery. The benefit that the system provides over conventional computer navigation is the addition of kinematic ligament balance data and then modification of the prosthesis placement based on the patient's anatomy, alignment [13], and soft tissue balance with joint obliquity [14]. This “personalized robotic execution” may be functionally beneficial and lead to earlier recovery. The robotic systems have the “execution” of the cuts within the system thus completing the “evolution” from navigation into robotics. The robotic systems cannot exist without some form of navigation in the system to verify the accuracy on the table. The robot either uses sculpting tools, burrs, or saws to provide the bone cuts.

Hip arthroplasty

This technique facilitates accuracy for implant placement [6] and perhaps more bone-conserving [15]. Autonomous robotic bone cutting involves the robot cutting bone without the assistance of a controlling human hand. Different types of robots are used in orthopaedic surgery: passive robots; robots with surgeon-guided systems called haptic semi-active robots; and active robots. Passive systems guide surgeons while they control robotic tooling to carry out a task. By limiting surgical manipulation through feedback, semi-active devices limit what can be done surgically. With the assistance of a whole-process robotic arm [4] specifically constructed, active systems may autonomously carry out activities without direct human manipulation and surgeon control intra-operatively. Robotic bone cutting can also be autonomous, haptic, or boundary, where human interaction is required to move the robot but cutting is disabled or prevented if it crosses a border even though it is free to move anywhere in space.

Spine surgery

Robotic spine surgery refers to robotic technology for guidance during spine orthopaedic surgery or neurosurgery. Traditionally, spine surgeons have inserted instruments “free-hand” into the spine, utilizing their anatomical knowledge

and X-rays. They must keep their hands stable while working close to sensitive nerves in small spaces, especially with today's minimally invasive procedures requiring considerably smaller incisions. By removing the issue of instability in surgeon's hands, robots enable surgeons to make precise movements while operating with challenging anatomy [16] for the placement of rods or screws [17, 18]. Robots can carry out repetitive activities for almost indefinite periods without experiencing performance degradation, which reduces the likelihood of fatigue-related mistakes and may lead to more reliable patient outcomes [19]. Radiation exposure for the patient, surgeon, and other operating room personnel [20] is also reduced when robotic guidance or other methods are used in place of fluoroscopic.

The recent introduction of artificial intelligence in orthopaedic surgery

Contrary to the introduction of robotics in surgery, the introduction of AI in orthopaedic surgery if we exclude robots is recent (less than 5 years). The number of publications regarding AI and surgery has increased dramatically over the last decade, particularly during the previous two years. This phenomenon can be explained by the exponential growth in computing power available to the largest AI training runs. In the literature, AI is classified in different ways as explained by Korneev et al. [21] and there is a certain degree of overlap among the different terms and definitions used. Some of the most notable and continuously evolving AI sub-domains are machine learning, artificial neural networks, computer



Fig. 1 Photograph of Philippe Hernigou



Fig. 2 Photograph of Sebastien Lustig



Fig. 3 photograph of Jacques Caton

vision, and natural language processing as described by the three authors of this Editorial (Figs. 1, 2 and 3).

Machine learning (ML)

Machine learning enables computers to learn and predict via pattern recognition. Standard computer programs are explicitly programmed with a precise purpose. However, ML enables a computer to learn by supervised learning (partial data labeling) and/or by detection of the data

structure itself (unsupervised learning) to explain/predict data without the need for explicit programming. ML has particular potential in detecting elusive (i.e., indiscernible to human analyses) patterns in large datasets. ML, through the use of nonlinear models (integration of multiple data sources), has been shown to be able to outperform logistic regression for predicting surgical situations. Compared to traditional statistical analysis, this approach identifies less direct and more sophisticated nonlinear relationships/multi-variate effects. De Villeneuve et al. [22] investigated the ability of a machine-learning mathematical model to calculate lower limb alignment. The authors utilized an algorithm built on a machine-learning technique to calculate angles automatically. The algorithm's reliability was assessed by evaluating the difference in determination precision between the surgeons and the algorithm. The algorithm demonstrated remarkable accuracy in automated angle measurement, enabling the calculation of limb frontal alignment to the nearest degree.

Artificial neural networks (ANNs)

A notable subdomain of ML, which takes its inspiration from the biological nervous system, is artificial neural networks (ANNs). These approaches use a neural network system with signal processing organized in layers of “neurons” (basic computational unit). In the clinical context, ANNs outscore more customary strategies for predicting risk. For example, according to Hernigou [23], the artificial neural networks could predict collapse in hip osteonecrosis within three years follow-up with 81.2% accuracy. The ANNs can also create tools such as a dislocation risk calculator [24] to help surgeons choose the best solution for their patients.

Computer Vision (CV) and augmented reality

This field pertains to the machine interpretation of videos, images, and sounds. Videos inherently have a massive amount of potential information: it has been estimated that 60 s of HD video has 25 times more data than an HD CT scan. Striking results have been achieved in object recognition, with human-like (or above-human) performance. These applications can provide enhanced augmented interpretation of data during image-guided surgical approaches in orthopaedic surgery as described in a review by Lex et al. [25], but also in trauma surgery [26]. This application of AI is not limited to vision but also can increase acoustic interpretations [27] during surgery, not only during arthroplasty as explained by Homma et al. [28] but also for spine surgery [20].

Natural Language Processing (NLP)

NLP is another important subdomain of AI aiming to create a machine that can comprehend human language. This technology can be used to scrutinize electronic medical records and identify specific statements or vocabulary in operative reports or in orthopaedic journals. This capability could be important for analyses of large volumes of medical-record data (specifically narrative documents). In some review analyses, screening a thousand publications is sometimes necessary due to the exponential increase of articles on Pubmed. The evaluation process can be enhanced using artificial intelligence. Natural Language Processing was used in this Special Issue by Hernigou et al. [24] for the extraction of hip dislocation data in systematic reviews by using the terms “disloc,” “sublux,” “reduc,” “reloc,” and “displace.”

Artificial intelligence and robots in the operating room: which future?

If knee and hip surgery were the first surgeries to benefit of robotic technology [29] Hand surgery [30] and pediatrics [31] are two different topics with possible future benefits of robotic surgery. As suggested by Lonner et al. [32] and Al Zoubi [33], robotic systems may also help for inventory, and may improve surgeon ergonomics by integrating algorithms in a large digital ecosystem. Finally, after hospital discharge, continuous/integrated/tailored monitoring by smart devices [7] might help follow-up (as keeping track of pain, range of motion after a surgical procedure). Moreover, AI will make global big data more accessible everywhere. Even after a long career and having managed several thousands of patients, a specialist will have “limited experience” compared to global cohorts of patients as explained in the paper evaluating of one million primary implants [24].

During the COVID period, healthcare has been the largest concern and, by any metric, the amount of money invested in health has increased by a substantial margin. After the COVID the widespread introduction of AI in healthcare with the growth in musculoskeletal pathology worldwide [34] can potentially increase the costs, but the productivity of the workflow, thus increasing efficiency also could reduce costs. At this stage, the actual full potential of this technology is still unclear. The outcome is challenging to quantify. Patients, hospital managers, and even surgeons (due to the costs of the system) may have very different expectations from this robotic surgery, as explained by two separate surveys on robotic knee surgery [35, 36].

Artificial intelligence and robotics limitations

Despite its vast applications in all the types of surgery [37], robots and AI are not a panacea for providing solutions to every question and problem. For robotics, without computer navigation, robotic systems cannot work as the accuracy cannot be guaranteed. As in previous navigation systems, if the registration process, or indeed the CT segmentation process is full of human errors, the execution performed by the robot will also be suboptimal. Furthermore, the surgeon has to decide the implant position based on further acquisition data (ligament balance and tension) acquired during surgery.

Despite the vast power of AI, there is an inherent limit related to data exactness (systematic bias, or an improper labeling/classification of data) which can hinder pattern recognition and the accuracy of the related predictions stemming from AI. In multiple domains, a classical analytical approach can still outperform the current level of AI. Despite the improvements in drawing causal inferences, the current level of AI is often unable to detect causal connections to a degree applicable in a clinical context, nor is it capable of automatic clinical interpretations. Despite its abundance of variables, the big data approach cannot provide the conceptual lens and the proper clinical framework to interpret the data.

The legal status of robots in the operating room

Generally speaking, technological objects—regardless of how basic or complex they may be—are regarded as tools or instruments of human decision-making and action. In addition to having the support of tradition, this instrumentalist concept has so far shown to be a helpful approach for addressing and comprehending advancements in robotics and artificial intelligence. However, we cannot exclude as orthopedic surgeons (knowing perfectly the difficulty of simple alignment of bones) that one of the risks for the surgeon could be the “misaligned” robot or the infection by a virus of the artificial intelligence of the computer in the operating room. This may create risks, depending on what systems we use and how we deploy them. Predicting the potential catastrophic impact of AI in the operating room is difficult. The potential scenarios, including events such as software viruses, may affect surgery but also anesthesiology. Accepting the idea of risk leads to the concept of responsibility.

The European Parliament [38] has suggested that robots might need to be considered as “electronic persons” for legal integration. This idea is related to an essential legal

question of responsibility met enthusiasm and resistance. Since a robot is typically dedicated to an implant of the corporate name, the status of the robot could be strictly considered as a legal embodiment to its manufacturing company. However, future “generic robots” might be more intelligent and support the implantation of arthroplasties from several manufacturers, and therefore rather depending of the surgeon. We could come back to a past paradigm, introducing a new category of “intelligent entity” between the patient and the surgeon, an “in-between status” with the risk of resemblance to ancient laws of slavery. The surgeon, the robot, and the patient who accepts the robot in such a strategy may have various responsibilities.

Conclusion

Many data need ethical issues, quality, interoperability, security, privacy, and property (health is one of the biggest data producers). Artificial intelligence and robots are the new companions of doctors and orthopedic surgeons. They require regulation and ethics within the platforms that maintain the algorithms with cyber-security for shared trust. The financialization of medicine (for example, radiology) raises the question of big-data ownership versus patients (from whom the data comes), doctors, and researchers (who produce them) because there can be resold.

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Code availability None.

Declarations

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Consent for publication The authors agree to publish the manuscript.

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